Afnan Habib

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Google Scholar

Profile

Artificial Intelligence and Data Science Master's student at Sejong University, specializing in Computer Vision, Explainable AI, and Large Language Models for medical imaging. Research Assistant at AISSLAB. Industry experience in developing and deploying computer vision, NLP, agentic AI, and multimodal multitask AI systems.

Education

M.Sc. Artificial Intelligence GPA: 4.45/4.5
Sejong University, Department of Artificial Intelligence, Seoul, South Korea March 2025 – February 2027
Thesis: A Generalist Multimodal and Multitask Medical CAD Framework with Modality-Task Conditioned Mixture-of-Experts

B.Sc. Telecommunication Engineering GPA: 3.67/4.0
University of Engineering Technology, Mardan, Pakistan September 2017 – August 2021
Thesis: 5G MIMO Antenna with AI-based Beamforming Capability For High Data Rate Base Station Communication

Experience

Research Experience

Research Assistant | AISSLab – Sejong University, Seoul, South Korea March 2025 – Present

- Conducting research on unified multimodal, multitask medical AI models across X-ray, CT, MRI, and ultrasound for segmentation, classification, detection, and clinical report generation.
- Developing vision-language models, LLM-based clinical reporting, Retrieval-Augmented Generation, and agentic AI systems using vector and graph databases, real-time retrieval, and MLOps/LLMOps tools.
- Publications in Artificial Intelligence Review, IEEE Journal of Biomedical and Health Informatics, IEEE BHI, MICCAI, and Applied Sciences.

Industry Experience - 2.5+ years of experience developing and deploying AI solutions.

Machine Learning Developer | Dcube Technologies – Islamabad, Pakistan Oct 2023 – Jun 2024

- Designed and implemented LLM-powered API automation system that translated natural-language instructions into schema-compliant JSON requests using GPT-3.5 and YAML-defined API specifications.
- Contributed across the development lifecycle, including data preparation, prompt and model refinement, continuous evaluation, error analysis, and Docker-based deployment to improve system reliability, portability, and production readiness.

Artificial Intelligence Developer | Genesis IT Lab Technologies – Islamabad, Pakistan Oct 2022 – Sep 2023

- End-to-end development of production AI solutions for real-time visual analytics, identity verification, customer engagement, and conversational automation, from data preparation and model evaluation to deployment.
- Built and optimized scalable inference pipelines for large-scale image and video workloads using PyTorch, TensorFlow, YOLOv7, InsightFace, ONNX, TensorRT, Triton Inference Server, Elasticsearch, and Docker.
- Improved the reliability and deployment efficiency of AI applications through systematic model benchmarking, latency optimization, containerization, and integration with backend services.

Patents

1. **A Generalist Multimodal and Multitask Medical CAD Framework with Modality-Task Conditioned Mixture-of-Experts** *Korean Patent*, **Under review**; submitted 2026.05.16

Journal Publications

1. Afnan Habib , Mugahed A. Al-Antari, Mukhlis Raza, Osamah F. Abdulmahmod, Hyunwook Kwon, Ertan Bütün, Metin Kaplan, Bilal Ertuğrul, Tülin Öztürk, and Yeong Hyeon Gu. **"A Unified Multimodal and Multitask Medical CAD Framework with Mixture of FreqXperts."** Submitted to the *IEEE Transactions on Medical Imaging* Special Issue on *Large Multimodal and World Models for Medical Imaging*. *Under Review*, 2026. *(Q1; Top 3% journal)*

2. Osamah F. Abdulmahmod, Hyunwook Kwon, Afnan Habib , Mukhlis Raza, et al. **"Medical Spine Sagittal MRI**

Dataset for Segmentation and Foraminal Stenosis Detection." *Scientific Data*, 2025. DOI: 10.1038/s41597-026-07138-x

3. Mukhlis Raza, Osamah F. Abdulmahmod, Afnan Habib , Hyunwook Kwon, Zaid Al-Huda, Omar Al-Maqtari, Ertan Bütün, Metin Kaplan, İsmail Akçin, Tülin Öztürk, Jamil Hussain, Yeong Hyeon Gu, and Mugahed A. Al-Antari*. **"From Images to Insights: Multi-Agent Conversational AI Framework for Multimodal Medical Imaging Analysis and Diagnosis."** *Alexandria Engineering Journal*, Elsevier. *Under Review*.

4. Saied Salem, Osamah F. Abdulmahmod, Afnan Habib , Mukhlis Raza, Hyunwook Kwon, Zaid Al-Huda, Omar Al-Maqtari, Ertan Bütün, Bilal Ertuğrul, Tülin Öztürk, Inbo Han, Yeong Hyeon Gu, and Mugahed A. Al-Antari. **"A 3D Vision-Language Framework for Automatic Lumbar Spine Diagnosis and Report Generation."** *Engineering Applications of Artificial Intelligence*. *Under Review*, 2025.

Conference Publications & Presentations

1. Afnan Habib , Mukhlis Raza, Osamah F. Abdulmahmod, Hyunwook Kwon, Ertan Bütün, Bilal Ertuğrul, Yeong Hyeon Gu, and Mugahed A. Al-Antari*. **"Med-GMoE: A Generalist Multimodal and Multitask Medical CAD Framework with Modality–Task Conditioned Mixture-of-Experts."** Submitted to the *International Conference on Medical Image Computing and Computer-Assisted Intervention (BHI 2026)*. *Under Review*.

2. Afnan Habib , Osamah F. Abdulmahmod, Mukhlis Raza, Yeong Hyeon Gu, Murat Aydoğan, and Mugahed A. Al-Antari. **"Towards Explainable AI in Agentic Retrieval-Augmented Generation: A Systematic Review."** In *Proceedings of the 9th International Artificial Intelligence and Data Processing Symposium (IDAP)*, Malatya, Türkiye, 2025. DOI: 10.1109/IDAP68205.2025.11222281

3. Mukhlis Raza, Saied Salem, Afnan Habib , Osamah F. Abdulmahmod, Hyunwook Kwon, Jamil Hussain, and Mugahed A. Al-Antari. **"MedXpert-CAD: A Multimodal Multi-Agentic System for Clinical Imaging Analysis via Model Context Protocol and LLM-Driven Agentic Workflows."** In *Agentic AI for Medicine, MICCAI 2025*, Lecture Notes in Computer Science, vol. 16147, Springer, 2025. DOI: 10.1007/978-3-032-06004-4_6. **Best Paper Award**

4. Saied Salem, Afnan Habib , Mukhlis Raza, Zaid Al-Huda, Omar Al-Maqtari, Bilal Ertuğrul, Özal Yıldırım, Yeong Hyeon Gu, and Mugahed A. Al-Antari. **"AutoSpineAI: Lightweight Multimodal CAD Framework for Lumbar Spine MRI Assessments."** In *Proceedings of the IEEE EMBS International Conference on Biomedical and Health Informatics (BHI)*, Atlanta, Georgia, USA, 2025. DOI: 10.1109/BHI67747.2025.11269503

5. Osamah F. Abdulmahmod, Afnan Habib , Yeong Hyeon Gu, M. Talo, Mukhlis Raza, and Mugahed A. Al-Antari. **"A Novel Lumbar Foraminal Stenosis Detection and Classification Framework Using a 3D Sagittal MRI Dataset."** In *Proceedings of the 9th International Artificial Intelligence and Data Processing Symposium (IDAP 2025)*, September 6–7, Malatya, Türkiye and Seoul, Republic of Korea.

6. Saied Salem, Afnan Habib , Mukhlis Raza, Ahmet Arif Aydin, Yeong Hyeon Gu, and Mugahed A. Al-Antari. **"SpineAutoCAD: Multimodal CAD System for Lumbar Spine MRI Analysis and Structured Report Generation."** In *Proceedings of the 9th International Artificial Intelligence and Data Processing Symposium (IDAP)*, Malatya, Türkiye, 2025. DOI: 10.1109/IDAP68205.2025.11222171

Projects

Unified Multimodal and Multitask Medical CAD Framework 2025 – Present

PyTorch, Swin Transformer, Mixture-of-Experts, DINO, Vision-Language Models, LLMs

- Designed an end-to-end medical AI framework that jointly performs image segmentation, disease classification, object detection, region-level severity assessment, and automated clinical report generation.
- Unified heterogeneous X-ray, CT, MRI, and ultrasound datasets through modality–task conditioned expert routing, enabling multiple imaging modalities and clinical tasks to be handled within a single model.

Smart Office Object Detection System - *Python, PyTorch, YOLO, Weights & Biases* 2025

- Developed an object-detection pipeline for identifying office equipment and environmental objects in a smart-office setting, including dataset preparation, model training, evaluation, and experiment tracking.
- Awarded **First Prize** in the HumbleBeeAI Hackathon Challenge, South Korea.

LLM-Based API Automation System | Dcube Technologies - *Python, OpenAI, REST APIs, Docker* 2023 – 2024

- Developed an LLM-powered system that translated natural-language instructions into schema-compliant JSON requests for automated API execution.
- Contributed to data preparation, prompt and model refinement, continuous evaluation, error analysis, and containerized deployment to improve reliability and production readiness.

Satisfaction of Things (SOT) | Genesis IT Lab Technologies - *PyTorch, TensorFlow, YOLOv7, InsightFace, ONNX, TensorRT, Triton, Elasticsearch, Docker* 2023

- Developed a real-time customer-feedback platform integrating gesture recognition, face recognition, demographic analysis, mask detection, pose estimation, person detection, and multi-object tracking.
- Optimized and deployed computer-vision models using ONNX, TensorRT, and Triton Inference Server, with WebSocket-based communication and Elasticsearch-backed data management.

Post-Interview Identity Verification System | Genesis IT Lab Technologies 2023

Python, Computer Vision, Face Recognition, Voice Analysis, Lip Synchronization, OCR

- Developed a multimodal identity-verification system for assessing candidate consistency across interview sessions using facial similarity, voice characteristics, lip synchronization, OCR-based name extraction, and eye-gaze analysis.
- Integrated outputs from multiple verification modules into structured PDF reports to support faster and more informed post-interview decision-making.

Funded Research Projects

- **An Explainable Hybrid AI Workflow for MRI Lumbar Stenosis Prediction**
  Project No. RS-2023-00256517, Funded by NRF, MICT, Korea. Role: Research Assistant.

Skills

Programming Languages: Python C++ MATLAB SQL

Machine Learning & Deep Learning: PyTorch TensorFlow Scikit-learn Hugging Face Transformers NumPy Pandas

Computer Vision & Medical AI: OpenCV YOLO Swin Transformer DINO Vision-Language Models Multimodal Learning Image Segmentation Object Detection

LLMs & Agentic AI: LangChain LangGraph AutoGen RAG Model Context Protocol Langflow n8n

MLOps & Deployment: Docker Git Weights & Biases ONNX TensorRT Triton Inference Server FastAPI Flask

Databases & Data Tools: MySQL PostgreSQL Redis Elasticsearch Neo4j Chroma CVAT Roboflow

Software Engineering: Object-Oriented Programming REST APIs Design Patterns Model Evaluation

Certificates

Hackathon Win Challenge HumbleBeeAI 2025, South Korea
Microsoft Technology Associate (MTA-981) - Python Programming
Deep Learning Specialization
Neural Networks and Deep Learning
100 Days of Code: The Complete Python Pro Bootcamp

Awards & Honors

- **Sejong University International Student RA Scholarship**, Fully Funded Master's Award (March 2025 – Feb 2027)
- **Student Talent Scholarship**, University of Engineering Technology, Mardan, Pakistan.

References

Dr. Mugahed A Al-Antari (Thesis supervisor)
Assistant Professor, AISSLab (Head): https://aisslab.com/
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